Vivek Sai

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Education

M.Sc. Computer Science (Autonomous Systems) | *University of Stuttgart*, Stuttgart, Germany — Oct 2024 – Present

B.Tech. Computer Science & Engineering (AI) | *Amrita Vishwa Vidyapeetham*, Chennai, India — Oct 2020 – Apr 2024

Work Experience

Working Student – Humanoid Robotics, Development & Validation — Apr 2026 – Present
Porsche Engineering Services GmbH — *Stuttgart, Germany*

- Developed and integrated robotics software for autonomous mobile platforms using ROS 2, NVIDIA Isaac ROS, and Jetson, enabling perception, localization, and navigation capabilities.
- Built and maintained Docker-based deployment workflows for embedded robotics applications, debugging complex middleware, networking, dependency, and containerization challenges across development and production environments.
- Integrated and evaluated hardware components including cameras, audio systems, and onboard compute, while contributing to simulation, visualization, and real-world robot validation.
- Collaborated with cross-functional engineering teams through code reviews, technical documentation, and feature validation, delivering production-ready robotics software and deployment improvements.

Student Assistant — Nov 2025 – Jun 2026
Socially Intelligent Robotics (SIR) Lab, University of Stuttgart — *Stuttgart, Germany*

- Performed complete hardware bring-up and configuration of 2 FER robotic platforms, enabling stable, reproducible operation across research experiments.
- Debugged hardware–software integration issues across sensor and control interfaces, cutting experiment setup failures by ∼40% and improving overall system uptime during trials.
- Prototyped and evaluated learning-based robotic behaviors in NVIDIA Isaac Sim, focusing on sim-to-real transfer.

Student Assistant — Dec 2024 – Mar 2026
Construction Robotics, University of Stuttgart — *Stuttgart, Germany*

- Developed ROS 2-based robotic software for manipulation tasks, including perception integration, task execution, and motion coordination.
- Implemented and validated pose estimation and control loops, analyzing failure cases across simulation and real robot deployments.
- Conducted sim-to-real validation across manipulation tasks, identifying a ∼15% perception accuracy gap between simulation and physical deployment and reducing it to under 5% through calibration and pipeline tuning.

Research Intern — Jun 2023 – Aug 2023
IIITDM Kancheepuram — *Chennai, India*

- Implemented and benchmarked deep learning models (CNNs, Vision Transformers) using PyTorch and TensorFlow.
- Designed data preprocessing, augmentation, and evaluation pipelines to analyze robustness and generalization of models.

Projects

ContactPilot — Vision-Based Robotic Grasping Pipeline | ***Python, PyTorch, MuJoCo, Open3D***

- Built an end-to-end pick-and-place pipeline for a Franka Panda arm, integrating Contact-GraspNet (PyTorch) for 6-DoF grasp prediction, MuJoCo physics simulation, and multi-camera point-cloud fusion from calibrated RGB-D input.
- Designed a seed-based benchmarking harness to systematically evaluate grasp performance, and iteratively tuned grasp ranking, gripper control, and contact-friction physics — raising pick-and-place success rate from 38% to 93%.

Vision-Language-Control Robot in Simulation | ***MuJoCo, Python***

- Developed a simulation-based robotic system combining state estimation, decision logic, and motion execution.
- Used simulation to prototype, debug, and validate perception-to-action pipelines in a controlled environment.

Differential Drive FusionBot: From CAD to Autonomous Mapping | ***ROS 2, Python***

- Designed a mobile robot from CAD to URDF, integrating sensors and control interfaces.
- Implemented Nav2-based path planning and AMCL localization in a ROS 2 architecture, achieving a 90% autonomous navigation success rate across repeated test runs.

Skills

Programming:	C++, Python, Linux/Bash
Robotics:	ROS 2, robot kinematics, motion execution, path planning, localization, sim-to-real transfer
Perception:	Intel RealSense D435i, NVIDIA cuVSLAM, nvblox, camera calibration, 6D pose estimation (AprilTag)
Simulation:	NVIDIA Isaac Sim, MuJoCo, Gazebo, URDF
ML / Deep Learning:	PyTorch, CNNs, Vision Transformers, reinforcement learning, model evaluation
Hardware:	Unitree Go2, Jetson Orin NX, FER robotic platforms
Tools:	Git, Docker, unitree_sdk2
Languages:	English (C1), German (A1)